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PI Name:	Crossin, John P M.S.		
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PI Email:	johnpcrossin@10blade.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	978-549-3868
Organization Name:	10 Blade, Inc.		
PI Address 1:	8 Sky Reach		
PI Address 2:			
PI Web Page:			
City:	Plymouth	**State:**	MA
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COI Name (Institution):	Beck, George (Impact Instruments) Moulton, Steven (10 Blade, Inc.) Hatfield, Thomas (Wyle Laboratories) Myung, Daniel (10 Blade, Inc.) Gaynor, Mark (Saint Louis University)		
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Task Description:

Overview: Astronauts on long-duration missions will not be able to return to Earth for routine or emergency medical care. We are developing an integrated system that can function autonomously, yet communicate with Earth to provide astronaut health assessment, maintenance, and medical care. The system is based on the integration of two existing medical platforms. iRevive, which is a semantically flexible, mobile electronic medical record, serves as the primary interface between the patient and crew medical officer. The lightweight trauma module (LTM), which is a combination ventilator/critical care monitor and therapeutic system with integrated automatic control systems, provides manifold physiological monitoring and support functions. The integrated system will collect, monitor, and fuse patient care information with physiological patient data to further study and optimize remote medical diagnosis, ventilatory support, IV (intravenous) fluid therapy, and treatment options. The heart of the system is a robust database with a user friendly graphical user interface (GUI). The former is backed by a rich data dictionary, while the latter simplifies data collection and data entry through a variety of pull-down menu options. The overriding goal is an intuitive, easy to use system that requires little medical training to understand and use.
Objectives:

1. Formalize communication protocol between iRevive and the LTM

2. Design GUI for LTM integration into clinical workflow

3. Alter iRevive GUI to accommodate LTM data fields

4. Re-factor iRevive Database

5. Test and integration

6. Clinical trial, system modifications, and data exchange

7. Prepare and submit written report.

Key Findings: During the second year of this program all objectives except data exchange were completed. Communication between the two systems is controlled by a firmware driven field-programmable gate array (FPGA) in the LTM. A protocol verification simulator was written, debugged, and used for testing with iRevive. All levels of communication, such as sensor definition, data definitions, transmission packets, alarms, and time synchronization were addressed and integrated into the simulator. Extensive work was done to define the data for nearly thirty vital signs delivered by approximately twenty sensor types. The data structure of the transport (UDP) packet is defined and implemented. Specific packets were defined, including types for synch packet, parametric vital sign packet, device alarm packet, wave data packet, and general LTM information. The structure for verification of the contents of each of these packets was implemented, as well as protocols for missed and lost packets. Internal controls were defined and resolved. Situations such as coordination of time between devices and resolution of redundant data such as pulse rate from different devices were addressed. As a result, communications data is presented in a concise, cohesive, and consistent manner. The resultant system is extensible and robust.

Impact of Key Findings: Data definitions within iRevive are in place, as is a new, browser based graphical user interface (GUI). The GUI is what allows the crew medical officer to enter all relevant data necessary to ensure an optimal outcome. The GUI directs the user to successive screens specific to an injury or condition. An example of this is a "body picker" pop-up. By simply touching a representation of the part of the body where the condition exists a series of screens appears to guide the entry of required data in a consistent, standardized manner. Successive screens are accessed until a complete description of the condition and its treatment is entered. Audio and visual alarms from the LTM are in place in iRevive to warn of impending problems or the need for immediate action. These alarms range from simple reminders to reassess the patient to more complex alarms concerning an abnormal set of vital signs or ventilator malfunction. All data collected is placed in a database that is flexible, reportable, queryable, and much faster than previous versions. Every action is time stamped; thus, a uniform record of actions, interventions, medications, observations, and vital sign data is collected and preserved for current medical and future research purposes. The final version of the iRevive - LTM integrated system was completed on budget and on time.

Proposed Research Plan: The LTM-iRevive system went through clinical testing at the C-STARS (Center for Sustainment of Trauma and Readiness Skills) laboratory at the University of Cincinnati in August 2011. Overall, the integrated system was well received. Future funding is required for clinical trials to improve the compatibility and usability of the combined system. The LTM should receive FDA (Food and Drug Administration) 510(k) approval in early 2012. Once that is in place the combined system could be rigorously tested in a variety of pre-hospital (ground and aeromedical transport) and bedside ICU settings. Feedback from these trials would be used to modify the screens and pull down menus. More importantly, the combined physiological waveform and clinical datasets collected during these trials could be used to support development of closed loop control algorithms for oxygenation, ventilation, and fluid management.

Rationale for HRP Directed Research:

Civilian pre-hospital providers (e.g., paramedics and emergency medical technicians) collect and act upon a wide variety of complex visual clues, while monitoring and adjusting to continually changing sets of vital signs. Pre-hospital vital sign information in the form of trending data is rarely integrated into the patient care record, so how the vital signs change in response to specific treatment measures is largely undocumented and poorly understood. As a consequence, the care that is provided to a patient in the field is anticipatory and reactive. It is not time sensitive and the accompanying patient medical record is oftentimes incomplete. Healthcare providers do not give medications or adjust the ventilator and IV fluids as often or as accurately as a smart, vigilant system might and, as a result, patients likely do not respond or recover as quickly as they could.
We have developed a fully integrated system for mobile critical care patient support and documentation. The iRevive-LTM system is composed of physiological sensors, monitors, and therapeutic hardware devices, linked by a suite of software applications. The components integral to the LTM include: a ventilator; 3/5/12-lead ECG (electrocardiogram); pulse oximeter; noninvasive blood pressure (NIBP); end-tidal carbon dioxide (EtCO2); patient temperature; invasive arterial and intracranial pressure monitoring capabilities; Ethernet communications; closed-loop control of oxygenation (and soon ventilation and IV fluid control); an integrated electronic medical record (iRevive) for clinical data storage and export; alarming, and smart help. The LTM supports up to three external intravenous (IV) pumps and is designed to support other to be developed noninvasive monitors, all connected via powered-USB ports. It

Research Impact/Earth Benefits:	supports several additional modules, including an oxygen concentrator, patient warming, and an anesthesia control module. Software will oversee a growing number of autonomous care applications within the integrated system, which will reduce the need for constant attention by a healthcare provider or crew medical officer. While space flight design requirements are of paramount importance to the current project, we are cognizant of U.S. military funding and civilian needs for improved transport monitoring technology. The small, lightweight, rugged, low power design specifications for space flight are equally important here on Earth. A transport monitor that goes into space should have facility for remote calibration and maintenance, as should a transport monitor that is deployed on the battlefield or in other remote locations. Incorporation of redundant systems, automated alarms and, increasingly, closed loop control algorithms will be essential. The value of consolidating patient monitoring, support, and documentation into a single system, capable of automatically collecting and transmitting real-time patient care data, cannot be overemphasized. Integrating these data streams has many advantages, not only in providing real-time information display both locally and centrally for triage decision support, but in trauma system development. More importantly, the physiologic and electronic patient care data that will be captured by the iRevive-LTM system will be fully integrated and time synchronized. New state-of-the-art machine learning, feature extraction, and advanced statistical methods are showing great promise in analyzing these types of complex data sets, uncovering many important, previously hidden physiological relationships and treatment effects. As these relationships are further defined and understood, our models of health and disease will become more complex and accurate. They will provide more reliable, real-time insight into the current and predicted future status of our patients. In time, machine-based comprehension of semantic clinical information together with real-time physiological data will lead to the development of fully autonomous patient care systems.
Task Progress:	Key to this system is the collection and presentation of data. This has required: 1) rewriting the iRevive GUI and database codebase using current technology; 2) defining the communication protocol between the LTM and iRevive; and 3) building a LTM simulator to debug the iRevive-LTM communication protocol, which was hardcoded into the LTM. In the past year we have rewritten the GUI and database for iRevive to increase the speed, portability, and maintainability of the codebase, which is now in Python for improved interaction with the Internet and the world wide web. The GUI has improved graphics and more complete pull down menus, to facilitate ease of use and generate a more complete record with less user training required. Another focus has been defining a communication protocol for the LTM to exchange data with iRevive. This protocol was implemented as an LTM simulator for testing, debugging, and then in a FPGA. Our approach defined a base protocol to transmit parametric vital sign information. It was then extended to transmit wave type data, alarms, device configuration, and connection initiation. This effort produced a high-level protocol specification. We are using traditional Internet protocols, including UDP at the transport layer to exchange end-to-end packets. This information includes a sequence number, time stamp, packet type, and meta information describing the data from the LTM. The packet format defines all parametric vital sign data from each device module in the LTM, including SpO2, EtCO2, temperature, invasive and non-invasive blood pressure, ECG, and ventilator settings. These packets contain meta information, which describes attributes such as when measurements should be flagged with an alarm and the measurement units. The packet format also defines device alarms including their priority, how an alarm is indicated to the user (i.e., audible and latching), and information unique to each LTM device module. Work was also done to define the protocols and processes required for waveform data as well as an initiation protocol. Waveform data will be displayed with allowable ranges. Initiation protocols will synchronize these ranges and timing as well as provide the connection between LTM and iRevive. As the funding period is now complete, and the defined goals reached, the technical, funding, and schedule risk for this proposal are by definition low or non-existent.
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